

Disseminated Intravascular Coagulation (DIC) Syndrome

DR\Abdelrahman Elzefzaf

Consultant in Intensive Care and Critical
Care

At Shebin El-Kom Specialized Surgery Hospital
and Ismailia Medical Complex

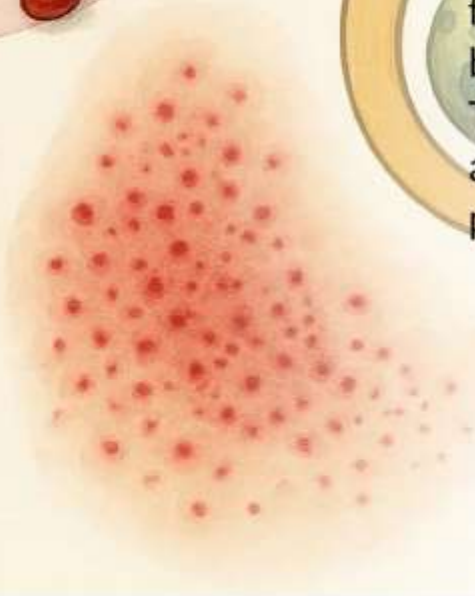
Master's Degree in Critical Care Egyptian
Fellowship in Intensive Care

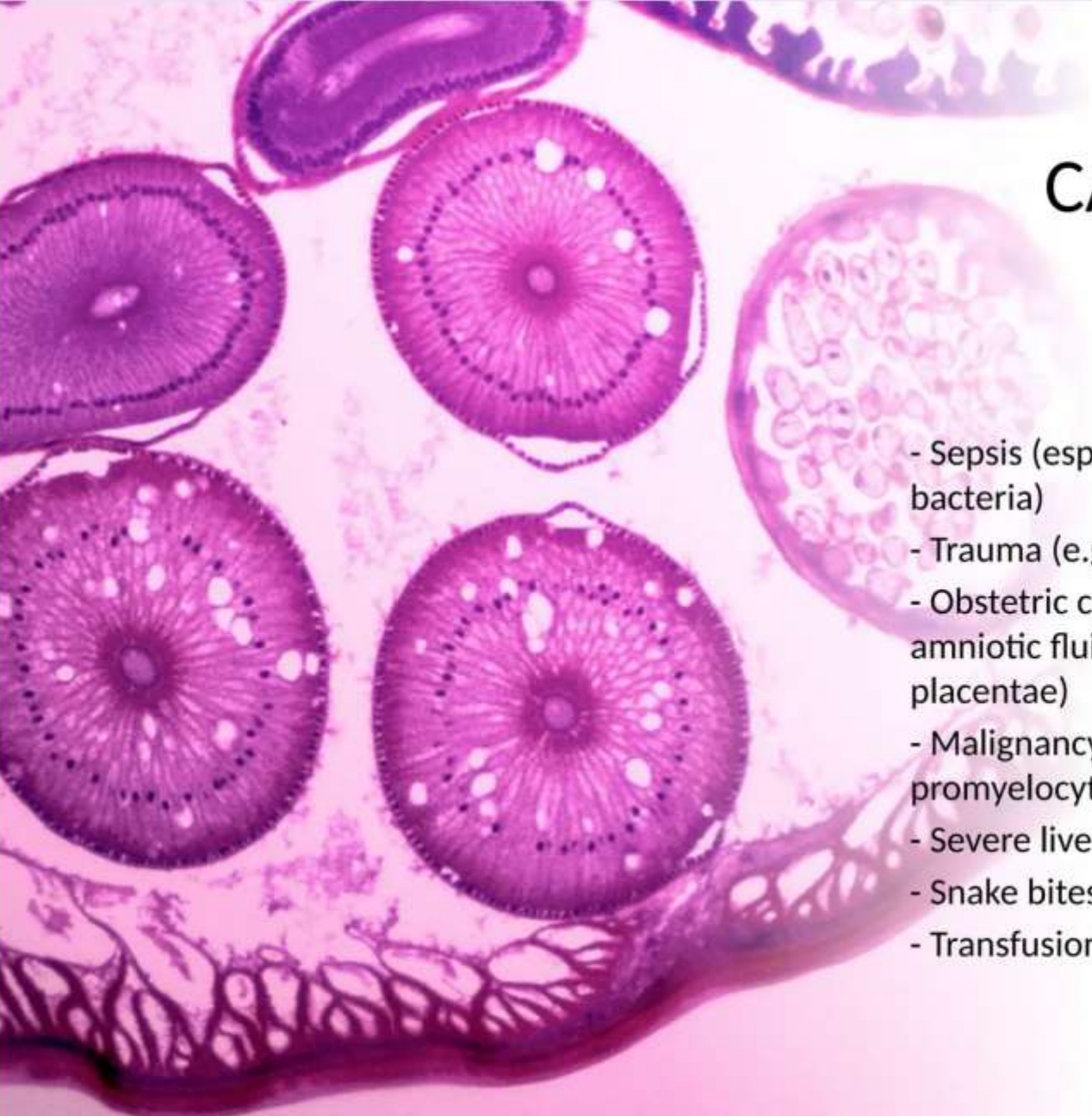




DEFINITION OF DIC

Disseminated Intravascular Coagulation (DIC) is a life-threatening condition characterized by systemic activation of blood coagulation, resulting in the formation of fibrin clots in small blood vessels throughout the body. This leads to multiple organ damage and depletion of clotting factors and platelets, causing serious bleeding.





CAUSES OF DIC

- Sepsis (especially gram-negative bacteria)
- Trauma (e.g., head injury, burns)
- Obstetric complications (e.g., amniotic fluid embolism, abruptio placentae)
- Malignancy (e.g., acute promyelocytic leukemia)
- Severe liver disease
- Snake bites
- Transfusion reactions

PATHOPHYSIOLOGY OF DIC

1. Triggering event activates the coagulation cascade.

2. Widespread microvascular thrombi formation.

3. Consumption of platelets and coagulation factors.

4. Activation of fibrinolysis leading to bleeding.

5. Result:
Thrombosis + Bleeding + Organ failure.

CLINICAL FEATURES OF DIC

- Bleeding (from venipuncture sites, mucosa, etc.)
- Petechiae and ecchymosis
- Digital ischemia or gangrene
- Hypotension and shock
- Signs of multiorgan dysfunction (renal, hepatic, CNS)



DIAGNOSIS OF DIC

- Low platelet count
- Prolonged PT and aPTT
- Low fibrinogen level
- Elevated D-dimer
- Presence of schistocytes on peripheral smear

Scoring systems like ISTH DIC score are used to aid diagnosis.



MANAGEMENT OF DIC

1. Identify and treat the underlying cause.

2. Supportive therapy:

Platelet transfusion if $<20,000/\mu\text{L}$ or if actively bleeding

Fresh frozen plasma (FFP) or cryoprecipitate for coagulation factor replacement

3. Heparin in selected cases where thrombosis predominates.



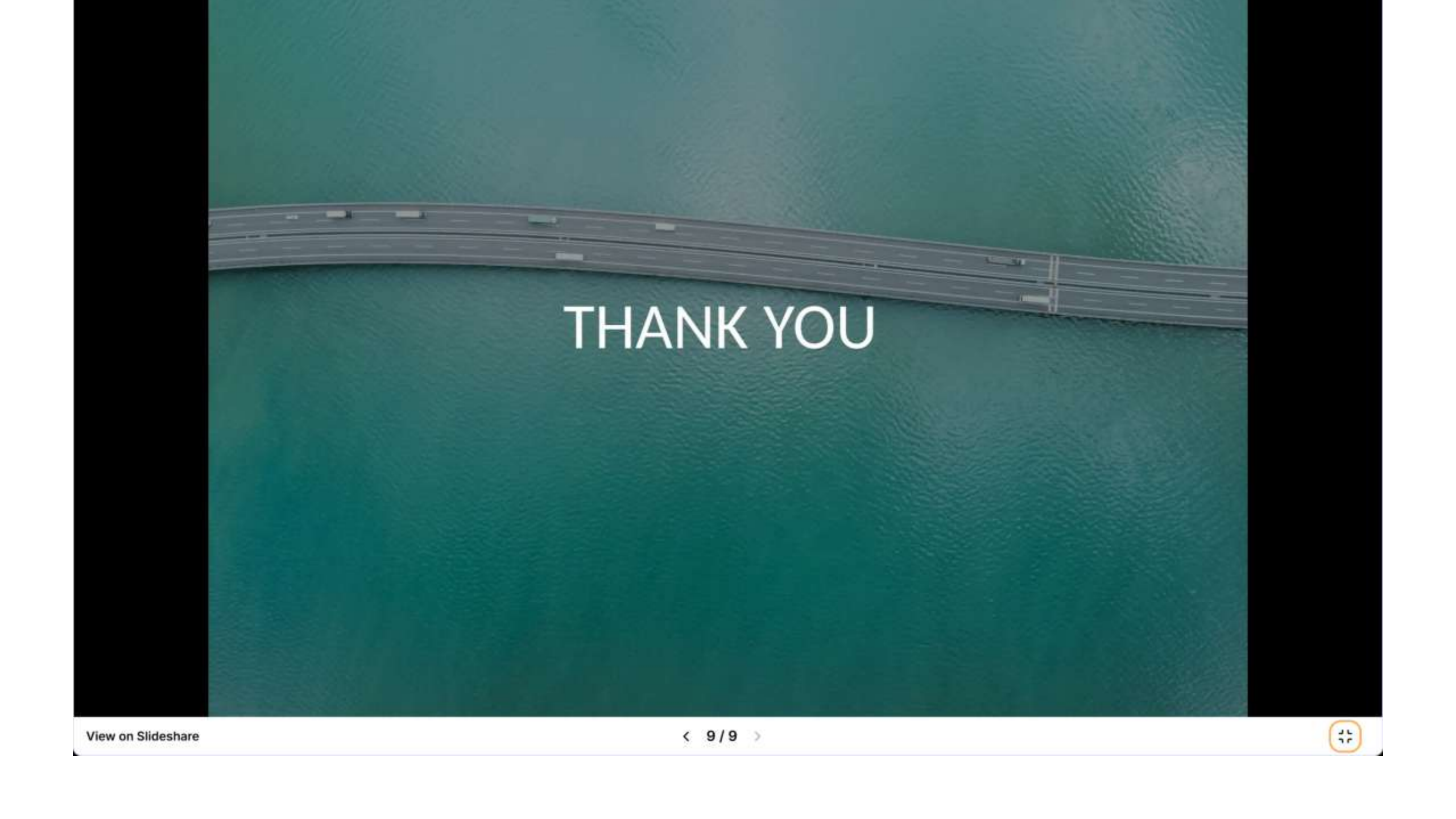
PROGNOSIS AND SUMMARY

- Prognosis depends on timely diagnosis and treatment of the underlying cause.

- DIC is not a primary disease but a syndrome secondary to other severe conditions.

- Prompt medical attention is crucial for patient survival.





THANK YOU

